

MOTION IN A STRIGHT LINE

Choose whether the statement is True (T) or False (F).

1. Path length is a scalar quantity, it has magnitude only and no direction. A scalar quantity can be negative also.
2. The magnitude of displacement may not be equal to path length traversed by the object. Magnitude of displacement may be greater than path length traversed.
3. Average speed does not tell in what direction object is moving but it tells about the type of path followed by the object.
4. Average speed may not be equal to magnitude of average velocity, and average speed during motion of the object is non zero.
5. For uniform motion, velocity is same as the average velocity at all instants and magnitude of instantaneous velocity is equal to instantaneous speed at that instant.
6. On a plot of velocity versus time, the average acceleration is the slope of straight line connecting points with initial velocity to final velocity.
7. Position-time graph with positive acceleration curves upward, downwards for negative acceleration and is a straight line for zero acceleration.
8. An object moving along negative direction on a straight line with negative acceleration is slowing down.
9. Area under curve for graph between velocity-time gives the distance covered over a given interval of time.
10. Acceleration and velocity cannot change values abruptly at an instant, changes are always continuous.
11. In case of constant acceleration, average velocity is equal to arithmetic mean of initial and final velocities.
12. The origin and positive direction of an axis in one dimensional motion is a matter of choice. You should first specify this point before you assign signs to quantities like displacement and velocity.
13. If a particle is speeding up along a straight line then acceleration can be either in same direction or in opposite direction of motion.
14. The zero velocity at any instant does not necessarily imply zero acceleration at that instant. A particle may be momentarily at rest and yet have non-zero acceleration.
15. The kinematic equations of motion are true only for motion in which magnitude and direction of acceleration are constant during the course of motion.
16. An object is released from great height and moves towards earth. If during fall, gravity disappears, object will stop its further motion.
17. An object in motion under constant acceleration always covers distance in ratio 1: 3: 5 in equal intervals of time.
18. An object thrown vertically upwards with speed greater than 9.8 m/s always cover a distance of 4.9 m during last second of upward journey.
19. Two objects are projected up from edge of a cliff with initial velocities v and $2v$. The graph showing time variation of relative position of second object with respect to first will be a straight line.
20. Two trains moving along straight tracks in same direction take less time to pass each other than to cross each other moving in opposite direction.

[illegible]

ANSWER KEY

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|----------------------------|----------------------------|
| 1. (T), [NCERT-XI, Pg.40] | 11. (T), [NCERT-XI, Pg.47] |
| 2. (F), [NCERT-XI, Pg.41] | 12. (T), [NCERT-XI, Pg.55] |
| 3. (F), [NCERT-XI, Pg.42] | 13. (F), [NCERT-XI, Pg.55] |
| 4. (T), [NCERT-XI, Pg.43] | 14. (T), [NCERT-XI, Pg.55] |
| 5. (T), [NCERT-XI, Pg.45] | 15. (T), [NCERT-XI, Pg.55] |
| 6. (T), [NCERT-XI, Pg.44] | 16. (F), [NCERT-XI, Pg.56] |
| 7. (T), [NCERT-XI, Pg.46] | 17. (F), [NCERT-XI, Pg.56] |
| 8. (F), [NCERT-XI, Pg.46] | 18. (T), [NCERT-XI, Pg.57] |
| 9. (F), [NCERT-XI, Pg.46] | 19. (F), [NCERT-XI, Pg.58] |
| 10. (T), [NCERT-XI, Pg.47] | 20. (F), [NCERT-XI, Pg.58] |



MOTION IN A PLANE

Choose whether the statement is True (T) or False (F).

1. A physical quantity is a vector quantity if it has direction and magnitude.
2. A vector $\vec{P}$ multiplied by a real number λ is also a vector, whose direction may be perpendicular to $\vec{P}$.
3. $\vec{C} + \vec{A} + \vec{B} \neq \vec{B} + \vec{A} + \vec{C}$
4. Motion in a plane can be treated as superposition of two independent simultaneous one dimensional motions along two perpendicular directions.
5. If a particle is moving in a plane and its speed is constant then distance traversed must be equal to displacement.
6. Equations of motion are applicable for uniform circular motion.
7. In non-uniform circular motion net acceleration of the body is towards center.
8. A scalar quantity is one that has the same value for observers with different orientation of axes.
9. If a vector is rotated its magnitude remains same.
10. The shape of the trajectory of the projectile motion under gravity depends on the initial conditions of motion also.
11. A particle executes a uniform circular motion and completes one revolution. Average velocity of the particle after completing of first quarter circle and first semicircle is same.
12. In ground to ground projectile motion, maximum height attained by projectile may be equal for two different angles of projection.
13. If two adjacent sides of a parallelogram are represented by $\vec{A}$ and $\vec{B}$ then $-\vec{C}$ will be along diagonal of parallelogram. Given that $\vec{A} + \vec{B} + \vec{C} = 0$.
14. In projectile motion under gravity, radius of curvature remains same in journey. ($\theta \neq 90^\circ$ and 0°)
15. The velocity vector of a particle at point is always along the tangent to the path of the particle at that point.
16. When a projectile is at the highest point of its trajectory, the momentum and acceleration are perpendicular to each other.
17. The acceleration vector of a particle in uniform circular motion averaged over one cycle is a null vector.
18. A particle is describing uniform circular motion such that its time period of revolution is T . Angle between average velocity in the interval $t = 0$ to $t = \frac{T}{4}$ and acceleration at $\frac{T}{4}$ is 45° .
19. Vectors $\vec{A} = 3\hat{i} + 4\hat{j}$ and $\vec{B} = 6\hat{i} + 8\hat{j}$ are parallel.
20. Resultant of three non-coplanar vectors may be zero.

[illegible]

ANSWER KEY

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|----------------------------|------------------------------------|
| 1. (F), [NCERT-XI, Pg.66] | 11. (F), [NCERT-XI, Pg.80] |
| 2. (F), [NCERT-XI, Pg.67] | 12. (T), [NCERT-XI, Pg.78] |
| 3. (F), [NCERT-XI, Pg.68] | 13. (T), [NCERT-XI, Pg.69] |
| 4. (T), [NCERT-XI, Pg.83] | 14. (F), [NCERT-XI, Pg.78, 81] |
| 5. (F), [NCERT-XI, Pg.85] | 15. (T), [NCERT-XI, Pg.87] |
| 6. (F), [NCERT-XI, Pg.85] | 16. (T), [NCERT-XI, Pg.78] |
| 7. (F), [NCERT-XI, Pg.85] | 17. (T), [NCERT-XI, Pg.87] |
| 8. (T), [NCERT-XI, Pg.87] | 18. (F), [NCERT-XI, Pg.73, 74, 87] |
| 9. (T), [NCERT-XI, Pg.88] | 19. (T), [NCERT-XI, Pg.67] |
| 10. (T), [NCERT-XI, Pg.85] | 20. (F), [NCERT-XI, Pg.67, 86] |



NEWTON'S LAWS OF MOTION

Choose whether the statement is True (T) or False (F).

1. A car sometimes overturns while taking a turn. When it overturns it is the inner wheel which leaves the ground first.
2. An elevator having mass m is pulled upwards with an acceleration a , then the tension in the cable is $m(g - a)$
3. A cricket ball of mass 25 g collides with a bat with velocity 10 m/s and returns with same speed within 0.01 second. The force acted on the bat is 50 N.
4. Swimming is possible on account of Newton's law of gravitation.
5. A man is standing on a weighing machine. If he jumps outside the machine, the reading of a weighing machine first increases then decreases to zero.
6. Force is required to keep the momentum of particle constant.
7. A block is kept on a frictionless inclined surface with an angle of inclination α . The acceleration of incline to keep the block stationary with respect to inclined plane is equal to $g \tan \alpha$.
8. The maximum speed that can be achieved without skidding by a car on a circular horizontal road is $\mu\sqrt{Rg}$, where μ is coefficient of static friction.
9. A vehicle of mass m is moving on a straight rough horizontal road with momentum p . If coefficient of friction between tyre and road be μ , then on applying brakes minimum stopping distance for vehicle is $\frac{p^2}{2\mu m^2 g}$
10. Starting from rest, a body slides down a 45° rough incline plane in twice the time it takes to slide down same smooth inclined plane. The coefficient of friction between body and the incline plane is 0.75.
11. A block starts moving on an incline plane of inclination θ of length L . The velocity of particle at the bottom of incline is $\sqrt{2gL(\sin \theta - \mu \cos \theta)}$, where μ is coefficient of friction.
12. It is easy to draw up a wooden block along a smooth incline plane than to haul it vertically, principally because the mass become smaller.
13. If a body of mass m falls vertically to the ground from height h_1 and rebound to a height h_2 , then change in momentum of the ball on striking the ground is $m(\sqrt{2gh_1} + \sqrt{2gh_2})$.
14. Total linear momentum of a system of particles is zero imply that kinetic energy of the system of particles is zero.
15. A car is moving on a curved road of radius R , banked at an angle θ . If coefficient of friction between the car and the road is μ_s , then maximum safe speed of car on this road is $\sqrt{\frac{gR(\mu_s + \tan \theta)}{(1 - \mu_s \tan \theta)}}$
16. One end of a string of length L is connected to a particle of mass m and other end is connected to a small peg on a smooth horizontal table. If particle moves in a circle with uniform speed v , then net force on the particle will be equal to T (Where T represents tension in the string).

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17. While walking on ice one should take small steps to avoid slipping.
18. A motorcycle is going on an overbridge of radius R . The driver maintains constant speed. As the motorcycle is ascending on the overbridge, the normal force on it increases.
19. Minimum acceleration, with which a fireman can slide down a rope when breaking strength of rope is $\frac{2}{3}$ of his weight, will be $\frac{g}{3}$.
20. A uniform rope of length L lies on a table. If coefficient of friction between table and rope is μ , then maximum length of the part of rope which can overhang from edge of table without sliding down will be $\frac{\mu L}{\mu+1}$



ANSWER KEY

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| 1. (T), [NCERT-XI, Pg.104] | 11. (T), [NCERT-XI, Pg.102] |
| 2. (F), [NCERT-XI, Pg.100] | 12. (F), [NCERT-XI, Pg.100] |
| 3. (T), [NCERT-XI, Pg.94] | 13. (T), [NCERT-XI, Pg.100] |
| 4. (F), [NCERT-XI, Pg.97] | 14. (F), [NCERT-XI, Pg.93] |
| 5. (T), [NCERT-XI, Pg.100] | 15. (T) |
| 6. (F), [NCERT-XI, Pg.95] | 16. (T) |
| 7. (T), [NCERT-XI, Pg.102] | 17. (T) |
| 8. (F), [NCERT-XI, Pg.104] | 18. (F) |
| 9. (T), [NCERT-XI, Pg.102] | 19. (T) |
| 10. (T), [NCERT-XI, Pg.102] | 20. (T) |

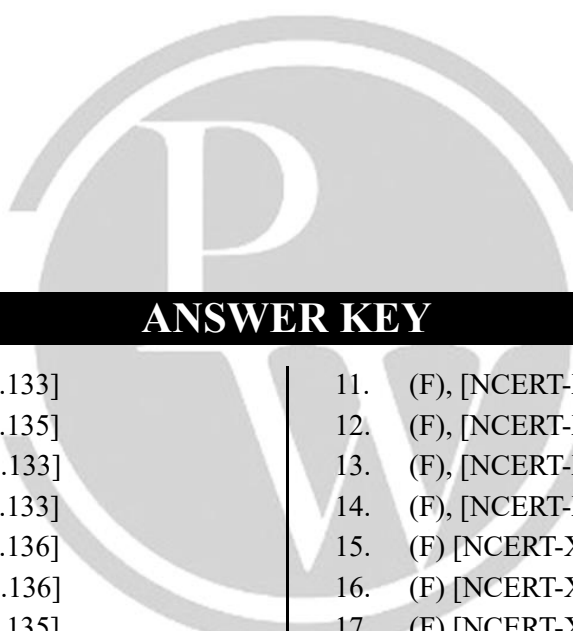


WORK ENERGY & POWER

Choose whether the statement is True (T) or False (F).

1. For two bodies, the sum of the mutual forces exerted between them (F_{12}, F_{21}) is zero from Newton's 3rd law, $F_{12} + F_{21} = 0$,
Sum of the work done by the two forces must be zero.
$$W_{12} + W_{21} = 0$$
2. In an elastic collision of two bodies, the momentum and energy of each body is conserved.
3. The principle of conservation of mechanical energy may be viewed as consequence of the work energy theorem for conservative forces.
4. The work energy theorem is independent of Newton's 2nd law.
5. In an elastic collision of two billiard balls, the total kinetic energy is conserved during the short time of collision of the balls (when they are in contact).
6. Total linear momentum is conserved during the short time of elastic collision of two balls (when they are in contact).
7. When a conservative force does positive work on a body, the potential energy of the body increases.
8. Work done by static friction on a body may be positive.
9. An artificial satellite orbiting the earth in very thin atmosphere loses its energy gradually due to dissipation against atmospheric resistance, however small. Its speed increases progressively as it comes closer and closer to the earth.
10. In an Inelastic collision, the total final kinetic energy of system is usually less than the total initial kinetic energy of the system.
11. $\vec{A} \cdot (\vec{B} + \vec{C}) \neq \vec{A} \cdot \vec{B} + \vec{A} \cdot \vec{C}$
12. Kinetic energy of a particle may be negative.
13. A particle is revolving on circular path under the influence of a force. Its velocity is changing then work done by force must be nonzero.
14. If work done by a force on a particle is negative, then kinetic energy of a particle is negative.
15. Work done by a man in lifting a bucket out of a well by means of a rope tied to the bucket is negative.
16. Absolute value of potential energy of a body subjected to a conservative force can be determined.
17. Total energy of a system is always conserved, no matter what internal and external forces on the system are present.
18. A body is dropped from a height H . The rate at which power is transferred by gravity is constant. (g is constant)
19. Total energy is conserved in inelastic collision.
20. In x-y plane two particles moving along x direction collides each other, inelastically. Linear momentum along y direction is not conserved.

[illegible]



ANSWER KEY

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|-----------------------------|-----------------------------|
| 1. (F), [NCERT-XI, Pg.133] | 11. (F), [NCERT-XI, Pg.132] |
| 2. (F), [NCERT-XI, Pg.135] | 12. (F), [NCERT-XI, Pg.117] |
| 3. (T), [NCERT-XI, Pg.133] | 13. (F), [NCERT-XI, Pg.117] |
| 4. (F), [NCERT-XI, Pg.133] | 14. (F), [NCERT-XI, Pg.116] |
| 5. (F), [NCERT-XI, Pg.136] | 15. (F) [NCERT-XI, Pg.134] |
| 6. (T), [NCERT-XI, Pg.136] | 16. (F) [NCERT-XI, Pg.120] |
| 7. (F), [NCERT-XI, Pg.135] | 17. (F) [NCERT-XI, Pg.135] |
| 8. (T), [NCERT-XI, Pg.117] | 18. (T) [NCERT-XI, Pg.128] |
| 9. (T), [NCERT-XI, Pg.135] | 19. (T) [NCERT-XI, Pg.129] |
| 10. (T), [NCERT-XI, Pg.135] | 20. (F) [NCERT-XI, Pg.131] |